Grid Distillation: Compositional Image Distillation via Structured Generative Grids

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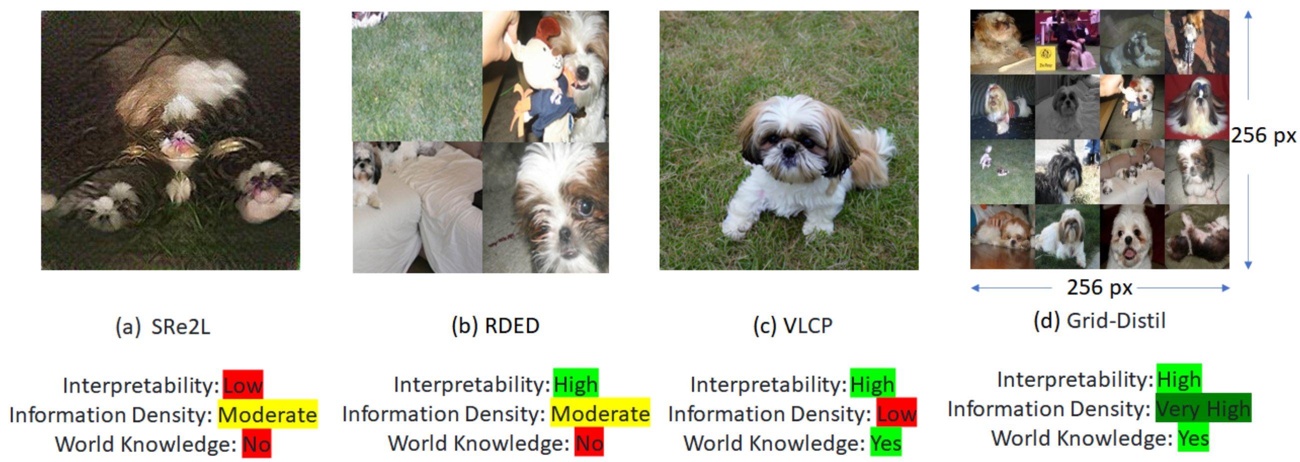

Figure 1. **Qualitative comparison of representations.** We visualize representative outputs from four paradigms: (a) SRe2L [43], (b) RDED [34], (c) VLCP [53], and (d) **Grid-Distil (ours)**. Each model captures different trade-offs between *interpretability*, *information density*, and *world knowledge*.

Abstract

*We present **Grid Distillation**, a generative dataset distillation framework that compresses large-scale datasets into a compact set of informative synthetic samples. Our method constructs high-resolution compositional grids via **spectral submodular optimization**, which injects world knowledge from CLIP representations to maximize semantic coverage and diversity. These grids are then downsampled into low-resolution distilled images optimized for diversity and representational efficiency. During training, a single-step diffusion reconstruction (based on Stable Diffusion Turbo) restores fine-grained spatial details from diffusion priors, bridging the gap between compact representations and natural image statistics. A grid-aware cropping strategy further enhances discriminability by probabilistically aligning crops with grid boundaries, maintaining compatibility with standard* 224×224 *inference inputs. Experiments on ImageWoof, ImageNette, ImageIDC, and ImageNet-1K demonstrate consistent improvements over existing dataset distillation methods across multiple IPC settings.*

1. Introduction

Deep learning has achieved remarkable progress across a broad range of computer vision tasks, primarily fueled by large-scale datasets and powerful computational resources [4, 14, 20]. However, the continuous expansion of data and model capacity has amplified storage, computation, and energy demands, raising concerns about reproducibility, sustainability, and accessibility. To mitigate these challenges, *dataset distillation* (DD) has emerged as a promising paradigm that aims to synthesize compact yet informative datasets capable of reproducing the performance of models trained on full-scale data [8, 12, 29, 39]. Such distilled surrogates significantly reduce dataset storage and computational overhead while providing benefits for privacy-preserving and copyright-conscious data sharing [1, 6, 16].

Early DD methods were primarily optimization-based, relying on meta-learning or distribution matching. Meta-learning approaches treat synthetic data as optimizable parameters within a bi-level optimization loop [5, 22, 23, 25, 33], while matching-based methods align gradients or intermediate feature statistics between real and synthetic samples [2, 12, 15, 19, 21, 37, 47, 49–51]. Although effective at small scales, these methods struggle with high-resolution datasets and large images-per-class (IPC) settings due to the prohibitive cost of iterative pixel-level optimization.

Recent advances in generative modeling have revitalized dataset distillation research by leveraging strong learned priors to synthesize realistic and diverse samples [9, 24, 32, 38]. By operating in latent spaces rather than directly on pixels, these generative DD approaches significantly improve fidelity and scalability. For instance, Gu et al.[9] introduce a minimax diffusion objective to enhance representativeness and diversity, while Su et al.[32] propose a clustering-based diffusion control strategy that aligns visual and textual prototypes for semantic consistency. However, despite these advances, existing diffusion-based distillation frameworks remain limited in how they encode spatial structure and compositional relationships.

Limitations of existing paradigms. Two complementary limitations characterize the current landscape. First, grid-like or patch-based condensation frameworks such as RDED [34] achieve efficient visual summarization but operate on disjoint cropped patches, which discard global spatial layout and contextual relationship inside the patch. This patch-centric design restricts the number of spatial units and leads to partial coverage of intra-class diversity. Second, methods such as VLCP [53](*Vision-Language Category Prototype*) leverage powerful diffusion priors to synthesize semantically rich samples but lack structured spatial composition—each image is generated independently, without explicit encoding of inter-instance or contextual dependencies. Consequently, both families of approaches fail to jointly capture *compositional structure* and *world knowledge*, resulting in distilled data that are either spatially fragmented or semantically shallow.

Our Approach. Our method constructs high-resolution compositional grids through *submodular selection* [13], ensuring maximal coverage and diversity while preserving the geometry of class manifolds, utilizing the world knowledge of CLIP model [28]. Each grid is subsequently downsampled to form compact **distilled images** optimized for representational efficiency.

To recover fine-grained spatial details lost during downsampling, we employ a **single-step diffusion reconstruction** inspired by InvSR [44]. InvSR introduced a diffusion inversion framework that restores high-frequency content using a noise-prediction model trained to approximate the reverse diffusion process in one step, thereby avoiding the inefficiency of multi-step sampling [11, 18, 40, 42, 45]. We adapt this fast diffusion prior for dataset distillation by prompting for grid specific target dataset, effectively utilizing a dataset-specific prior that reconstructs spatial details while injecting *world knowledge* from pretrained diffusion backbones. This one-step reconstruction bridges the gap between efficiency and fidelity-preserving global compositional structure from grids and restoring lost high-frequency details during distillation.

Finally, we introduce a **grid-aware cropping** strategy that probabilistically biases sampling toward grid-aligned regions, enhancing discriminability while maintaining test-time compatibility with standard $224{\times}224$ inputs. Together, these innovations yield a unified framework that leverages structured selection and fast diffusion priors for high-quality, semantically faithful, and spatially complete dataset distillation.

Contributions. Our main contributions are summarized as follows:

- We propose **Grid Distillation**, a unified dataset distillation framework combining submodular selection via our proposed Spectral Submodular Differentiable Information Maximization (SSDIM) and diffusion priors to encode intra-class diversity and compositional structure.
- We adapt a **Fast single-step diffusion reconstruction** mechanism, inspired by InvSR [44], tailored for restoring fine-grained spatial details lost during dataset compression.
- We propose a **Grid-aware Cropping** strategy that leverages structured layouts to improve robustness and semantic discrimination.
- Extensive experiments on ImageWoof, ImageNette, ImageIDC, and ImageNet-1K show consistent performance gains over distillation baselines across diverse IPC settings.

2. Methodology

We propose **Grid Distillation (GD)**, a compositional framework that distills compact yet expressive datasets by leveraging the spatial structure of image grids. Instead of synthesizing individual samples, GD generates grid layouts that encode intra-class diversity and contextual coherence within a single image.

2.1. Spectral-Submodular Grid Selection

A key objective of **Grid Distillation** is to compress the semantic and visual diversity of a class into a single $L \times L$ grid that maximally preserves its representative structure. Unlike random sampling or clustering-based aggregation, which often overrepresent dense appearance modes and ignore rare but informative variations, we cast grid construction as a principled *submodular selection problem* defined over image embeddings. This formulation enables a balanced selection of images that are simultaneously representative, diverse, and geometrically aligned with the intrinsic structure of the class manifold.

Embedding and affinity construction. Given M images belonging to a class, we extract their normalized feature embeddings $\{\mathbf{e}_i\}_{i=1}^M$ using a pretrained vision encoder (e.g.,

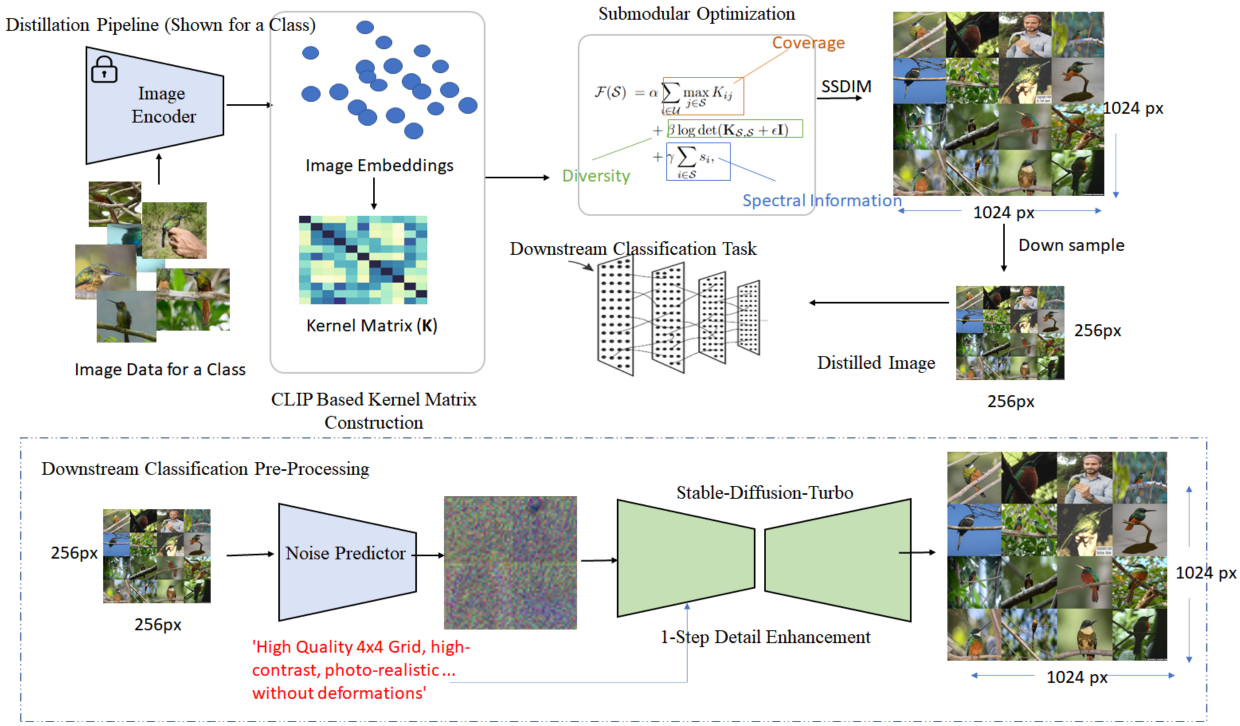

Figure 2. Spectral-submodular grid distillation pipeline. Starting from a pool of class images, we (1) extract normalized embeddings using a pretrained encoder, (2) build an affinity kernel and compute a spectral decomposition to obtain per-sample spectral energy scores s_i, and (3) solve a submodular optimization that combines coverage, diversity (log-det), and spectral informativeness to select L^2 exemplars which are down-sampled to get a distilled image. The distilled images are used in downstream classification task by applying single step diffusion to enhance lost spatial details.

CLIP [28]), such that $\|\mathbf{e}_i\|_2 = 1$. We define an affinity kernel $\mathbf{K} \in \mathbb{R}^{M\times M}$ with entries $K_{ij} = \mathbf{e}_i^\top \mathbf{e}_j$, measuring pairwise similarity in the embedding space. This kernel captures both visual and semantic relations among samples, serving as the foundation for subsequent manifold and submodular analysis.

Spectral manifold representation. To capture the latent geometry of the class, we perform spectral decomposition of the affinity matrix:

$$\mathbf{K} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top, \tag{1}$$

where $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_r]$ contains the top-r eigenvectors and $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_r)$ holds their corresponding eigenvalues. These spectral components characterize dominant modes of intra-class variation—such as pose, texture, viewpoint, and illumination—offering a compact and interpretable basis for selection.

For each image i, we define its *spectral energy score*

$$s_i = \sum_{k=1}^{r} \lambda_k\, u_{ik}^2, \tag{2}$$

which measures the total eigen-energy contributed by image i to the top-r manifold modes. Intuitively, s_i quantifies how structurally informative an image is with respect to the class manifold: higher values correspond to images that lie on high-variance regions or capture distinctive semantic configurations. The cumulative spectral energy of a subset $\mathcal{S}$ is then

$$E(\mathcal{S}) = \sum_{i\in\mathcal{S}} s_i = \mathrm{Tr}\big(\mathbf{U}_{\mathcal{S},:}\mathbf{\Lambda}\mathbf{U}_{\mathcal{S},:}^\top\big), \tag{3}$$

where $\mathbf{U}_{\mathcal{S},:}$ denotes the rows of $\mathbf{U}$ indexed by $\mathcal{S}$.

Submodular Optimization Objective. We seek a subset $\mathcal{S} \subseteq \{1, \ldots, M\}$ with $|\mathcal{S}| = L^2$ that jointly maximizes representativeness, diversity, and spectral informativeness:

$$\begin{aligned}\mathcal{F}(\mathcal{S}) &= \alpha \sum_{i\in\mathcal{U}} \max_{j\in\mathcal{S}} K_{ij} \\ &\quad + \beta \log\det(\mathbf{K}_{\mathcal{S},\mathcal{S}} + \epsilon\mathbf{I}) \\ &\quad + \gamma \sum_{i\in\mathcal{S}} s_i,\end{aligned} \tag{4}$$

where $\mathcal{U}$ denotes all candidate images and $\mathbf{K}_{\mathcal{S},\mathcal{S}}$ is the kernel submatrix induced by $\mathcal{S}$. The three terms respectively capture complementary desiderata:

- **Coverage** (α): Ensures that every image in $\mathcal{U}$ is well represented by at least one selected exemplar, promoting representativeness across the class distribution.
- **Diversity** (β): Encourages the selected embeddings to span a large volume in feature space, similar to a determinantal point process (DPP), thus avoiding redundancy.

- **Spectral Information** (γ): Biases selection toward high-energy samples that contribute strongly to the dominant manifold directions, aligning the grid with intrinsic semantic variations.

Together, these criteria form a balanced, theoretically grounded objective that naturally favors both visual coverage and manifold consistency—two properties often at odds in purely clustering-based methods. We apply this iteratively by removing the used candidates till we get required number of Grids, G or exhaust the candidate pool.

Spectral Submodular Differentiable Information Maximization (SSDIM).

Direct combinatorial optimization of Eq. 4 over the entire candidate pool $\mathcal{U}$ is computationally intractable for large-scale datasets. To address this, we propose **SSDIM**—a hybrid optimization framework that integrates differentiable submodular relaxation, spectral regularization, and information-theoretic diversity. SSDIM operates in three stages: (i) *Continuous relaxation*, where discrete selection indicators are projected onto a differentiable simplex via a Frank–Wolfe–style update, enabling gradient-based optimization; (ii) *Spectral regularization*, where eigenvalue-weighted feature embeddings promote coverage of dominant spectral modes of the affinity matrix; and (iii) *Greedy refinement*, which iteratively rounds fractional solutions using marginal gains from the submodular objective $\mathcal{F}$, ensuring monotone improvement and diversity consistency. This procedure yields compact yet representative grid selections that balance global coverage and local complementarity while remaining computationally tractable (Listed as Algorithm 1).

Resulting grids. The final L^2-image subset $\mathcal{S}$, obtained via SSDIM, forms a grid that encapsulates maximal intra-class variability and semantic completeness. Each grid acts as a distilled summary of its class—compressing diverse visual modes into a single structured image—and can be directly used as a generative conditioning cue or as a compact dataset element for downstream training.

Algorithm 1: Spectral Submodular Differentiable Information Maximization (SSDIM)

Input: Affinity matrix $\mathbf{K}$, spectral scores $\mathbf{s}$, grid size L,
weights α, β, γ, max iterations T

Output: Selected grid set $\mathcal{S}$ with $|\mathcal{S}| = L^2$

Initialization: Initialize continuous selection vector $\mathbf{p} \in [0,1]^M$ uniformly with $\sum_i p_i = L^2$; Initialize $\mathcal{S} = \emptyset$.

for $t = 1$ **to** T **do**

```
// 1. Continuous relaxation via Frank-Wolfe
```

Compute gradient: $\nabla_{\mathbf{p}}\tilde{\mathcal{F}}(\mathbf{p}) = \alpha\mathbf{K}\mathbf{p} + \beta(\mathbf{K}_{\mathcal{S},\mathcal{S}} + \epsilon\mathbf{I})^{-1}\mathbf{K}_{\mathcal{S},:} + \gamma\mathbf{s}$;

Find vertex direction: $v = \arg\max_{e_i}\langle e_i, \nabla_{\mathbf{p}}\tilde{\mathcal{F}}(\mathbf{p})\rangle$;

Update: $\mathbf{p} \leftarrow (1-\eta_t)\mathbf{p} + \eta_t v$, with step size $\eta_t = \frac{2}{t+2}$;

```
// 2. Rounding to discrete subset
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Select top-L^2 indices with largest p_i values as initial set $\mathcal{S}$.

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// 3. Local greedy refinement
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repeat

For each candidate $x \notin \mathcal{S}$ and $y \in \mathcal{S}$: compute marginal gain $\Delta_{x,y} = \mathcal{F}((\mathcal{S} \setminus \{y\}) \cup \{x\}) - \mathcal{F}(\mathcal{S})$; If $\max_{x,y}\Delta_{x,y} > 0$, apply swap $\mathcal{S} \leftarrow (\mathcal{S} \setminus \{y\}) \cup \{x\}$.

until *no improvement*;

return $\mathcal{S}$

2.2. Spatial Detail Reconstruction by Diffusion Inversion

To enhance the visual fidelity and semantic alignment of the distilled grids, we adapt the InvSR process [44] to a grid setting. Similar to super-resolution, where the objective is to recover spatial detail from low-resolution inputs, our goal is to reconstruct semantically coherent high-frequency textures within each grid cell while preserving inter-cell compositional structure.

Given a submodularly selected high-level grid $\boldsymbol{G}$ containing L^2 image patches, we downsample it to a compact distilled representation $\boldsymbol{y}_0$ that captures coarse features and semantics. During, downstream training, a noise prediction model f_w then predicts a noise map conditioned on both $\boldsymbol{y}_0$ and a text or class embedding $\boldsymbol{p}$ derived from CLIP. This prompt-conditioning encourages the diffusion model to synthesize details that are not merely perceptually plausible but also semantically aligned with the class concept. The initial latent state for inversion is thus formulated as

$$\boldsymbol{x}_{\tau_S} = \sqrt{\bar{\alpha}_{\tau_S}}\boldsymbol{y}_0 + \sqrt{1-\bar{\alpha}_{\tau_S}}f_w(\boldsymbol{y}_0, \boldsymbol{p}, \tau_S), \tag{5}$$

where τ_S denotes the starting diffusion step. A single-step or few-step reverse diffusion process (e.g., Stable Diffusion Turbo) is then applied to reconstruct the refined grid $\boldsymbol{x}_0$:

$$\boldsymbol{x}_0 = g_\theta(\boldsymbol{x}_{\tau_S}, \tau_S), \tag{6}$$

yielding grid images with enhanced realism and semantic richness.

In contrast to InvSR [44], which reconstructs natural HR images, our adaptation leverages the compositional structure of grids and text-conditioned priors to generate *grid-consistent manifold refinements*. This allows diffusion pri-

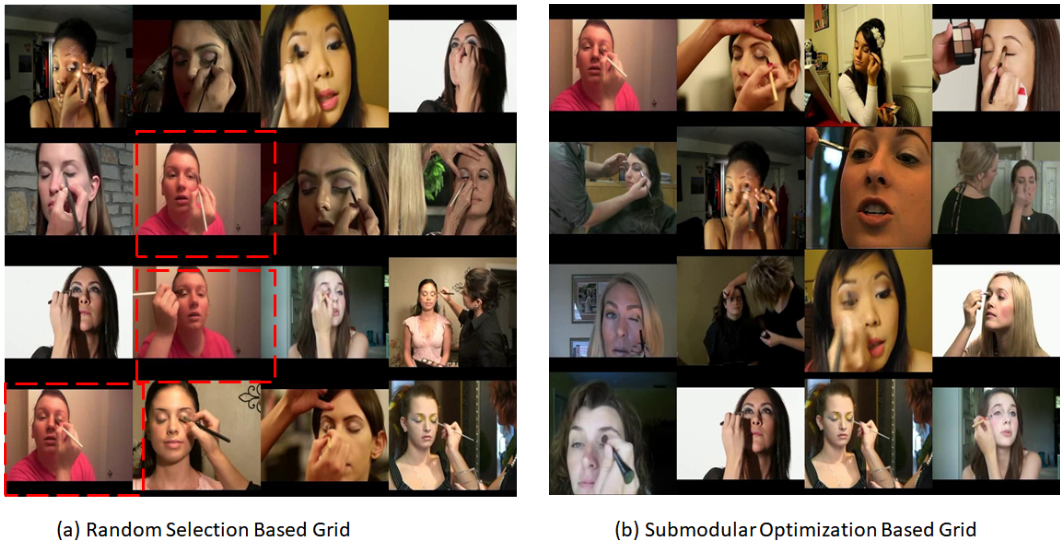

Figure 3. **Effect of Submodular Grid Selection.** Comparison of grids constructed from the UCF Mid-Frames dataset (selected to show the visible effect of our algorithm). **Left:** Random selection often yields redundant sub-images (highlighted in red). **Right:** Our spectral submodular optimization produces semantically diverse and spatially complementary grids. The figure demonstrates how submodular selection improves coverage and diversity over naive random sampling.

ors to inject world knowledge in a controllable and class-aware manner while maintaining the efficiency of partial noise prediction. Empirically, we found that a a single inversion step ($S = 1$) suffices for stable and high-fidelity grid reconstruction, taking approx. 148ms. However, the noise prediction model on A6000 GPU takes about 918ms which can be computed once per image.

2.3. Grid-Aware Cropping Strategy

To leverage the compositional structure of spatially enhanced distilled grids during training, we introduce a **grid-aware cropping** transform that mixes structured and random sampling. Given a grid image $\mathbf{I} \in \mathbb{R}^{H\times W}$ composed of L^2 cells of size $h \times w$, the crop operator $\mathcal{C}(\mathbf{I}; p_{\text{align}})$ selects a patch as:

$$\mathcal{C}(\mathbf{I}; p_{\text{align}}) = \begin{cases} \text{AlignedCrop}(\mathbf{I}), & \text{w.p. } p_{\text{align}}, \\ \text{RandomCrop}(\mathbf{I}), & \text{w.p. } 1 - p_{\text{align}}. \end{cases} \tag{7}$$

An *aligned crop* starts at multiples of (h, w), preserving the grid's semantic layout; a *random crop* samples arbitrary offsets, adding local variation. The resulting crops preserve semantic coherence within grid cells while promoting robustness to off-grid perturbations, ensuring compatibility with standard 224×224 test-time inputs.

3. Experiments

3.1. Datasets

We evaluate our proposed **Grid Distillation** framework on both distillation evaluation datasets publised in the the prior works. For high-resolution experiments, we consider subsets of ImageNet-1K [4]: ImageWoof, ImageNette [10], ImageIDC and [12]. ImageWoof is a challenging subset with 10 fine-grained dog breed classes exhibiting high inter-class similarity. ImageNette and ImageIDC also contain 10 classes but with lower inter-class similarity, making them comparatively easier. ImageIDC is derived from first 10 classes the ImageNet-100 [35] which contains 100 classes.

For action recognition, we evaluate on UCF-MidFrames, which contains video frames from multiple action categories. Following LDC [17], we adopt few shot classification metric for action classification, distinct from the image classification Minimax protocol [9].

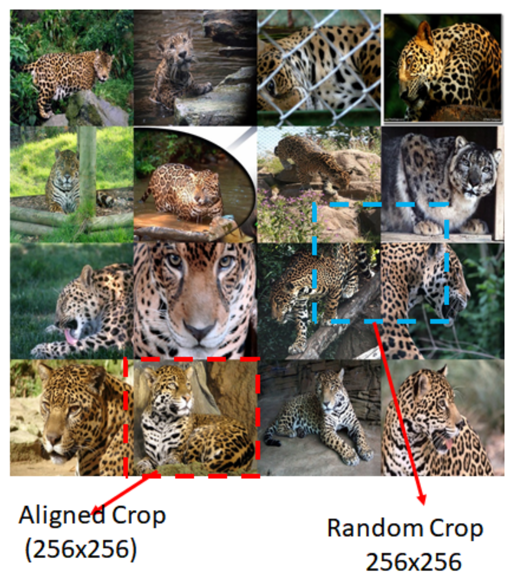

Figure 4. **Illustration of Grid-Aware Cropping** An *aligned crop* starts at multiples of (h, w), preserving grid's semantic layout; a *random crop* samples arbitrary offsets, adding local variation.

IPC (Ratio)	Test Model	Random	K-Center	Herding	DiT	DM	IDC-1	GLaD	Minimax	D4M	VLCP	Ours
10 (0.8%)	ConvNet-6	24.3±1.1	19.4±0.9	26.7±0.5	34.2±1.1	26.9±1.2	33.3±1.1	33.8±0.9	33.3±1.7	29.4±0.9	34.8±2.4	**49.7±0.7**
	ResNetAP-10	29.4±0.8	22.1±0.1	32.0±0.3	34.7±0.5	30.3±1.2	39.1±0.5	32.9±0.9	36.2±3.2	33.2±2.1	39.5±1.5	**63.9±0.6**
	ResNet-18	27.7±0.9	21.1±0.4	30.2±1.2	34.7±0.4	33.4±0.7	37.3±0.2	31.7±0.8	35.7±1.6	32.3±1.2	39.9±2.6	**65.5±0.8**
20 (1.6%)	ConvNet-6	29.1±0.7	21.5±0.8	29.5±0.3	36.1±0.8	29.9±1.0	35.5±0.8	-	37.3±0.1	34.0±2.3	37.9±1.9	**63.0±1.7**
	ResNetAP-10	32.7±0.4	25.1±0.7	34.9±0.1	41.1±0.8	35.2±0.6	43.4±0.3	-	43.3±2.7	40.1±1.6	44.5±2.2	**73.7±0.7**
	ResNet-18	29.7±0.5	23.6±0.3	32.2±0.6	40.5±0.5	29.8±1.7	38.6±0.2	-	41.8±1.9	38.4±1.1	44.5±2.0	**75.2±0.6**
50 (3.8%)	ConvNet-6	41.3±0.6	36.5±1.0	40.3±0.7	46.5±0.8	44.4±1.0	43.9±1.2	-	50.9±0.8	47.4±0.9	54.5±0.6	**79.0±0.6**
	ResNetAP-10	47.2±1.3	40.6±0.4	49.1±0.7	49.3±0.2	47.1±1.1	48.3±1.0	-	53.9±0.7	51.7±3.2	57.3±0.5	**82.4±0.2**
	ResNet-18	47.9±1.8	39.6±1.0	48.3±1.2	50.1±0.5	46.2±0.6	48.3±0.8	-	53.7±0.6	53.7±2.2	58.9±1.5	**84.3±0.9**
70 (5.4%)	ConvNet-6	46.3±0.6	38.6±0.7	46.2±0.6	50.1±1.2	47.5±0.8	48.9±0.7	-	51.3±0.6	50.5±0.4	55.8±1.7	**80.1±0.8**
	ResNetAP-10	50.8±0.6	45.9±1.5	53.4±1.4	54.3±0.9	51.7±0.8	52.8±1.8	-	57.0±0.2	54.7±1.6	60.6±0.3	**81.8±0.6**
	ResNet-18	52.1±1.0	44.6±1.1	49.7±0.8	51.5±1.0	51.9±0.8	51.1±1.7	-	56.5±0.8	56.3±1.8	60.3±0.3	**85.5±0.9**
100 (7.7%)	ConvNet-6	52.2±0.4	45.1±0.5	54.4±1.1	53.4±0.3	55.0±1.3	53.2±0.9	-	57.8±0.9	57.9±1.5	62.7±1.4	**80.7±0.6**
	ResNetAP-10	59.4±1.0	54.8±0.2	61.7±0.9	58.3±0.8	56.4±0.8	56.1±0.9	-	62.7±1.4	59.5±1.8	65.7±0.5	**83.4±0.7**
	ResNet-18	61.5±1.3	50.4±0.4	59.3±0.7	58.9±1.3	60.2±1.0	58.3±1.2	-	62.7±0.4	63.8±1.3	68.3±0.4	**85.9±0.2**

Table 1. Comparison of state-of-the-art methods on ImageWoof under various IPC settings and model architectures. "Ours" column reports accuracies from our Grid-Distil experiments. All results are measured at 256×256 resolution. The best results are in bold, and the second-best are underlined (in VLCP).

3.2. Implementation Details

Evaluation Setup: We utilize a single A6000 GPU with 48GB memory for performing all the experiemnts. All experiments for ImageNet-1K subsets are repeated for three independent trials with different random seeds, and mean ± standard deviation is reported. For ImageNet-1K subsets, the image resolution is set to 256×256 following MiniMax Protocol, and 224×224 for full ImageNet-1K following RDED Protocol. UCF-101 Midframes experiments are conducted at 256×256 resolution. For, UCF-101[31] Midframes a batch size of 16 is used for fine-tuning over 50 epochs as per implementation shared by LDC[17]. For action dataset, frames are treated as images, and the same Grid Distillation pipeline is applied with minor adaptations to integrate to few shot pipeline.

For fair comparison, we use publicly available implementations for baseline methods including Minimax [9], D^4M [32], GLaD [3], DiT [9, 27], IDC-1 [12], DM [48], SRe^2L [43], RDED [34], Herding [41], and K-Center [30].

Implementation Details of SSDIM. We implement the **SSDIM** module using openai clip implentation with batched GPU operations for efficiency. We utilize batch size of 64 to compute the embeddings. For an imagenet class, with approx. 1,300 images to construct the similarity kernel matrix, it requires approximately **18 seconds**. The subsequent grid generation stage synthesizes **10 enhanced grids per class** in about **57 seconds**, including function optimization and refinement using our non-optimized numpy implementation. All timings are measured on a single NVIDIA A6000 GPU.

Hyperparameter specification. Submodular weights $\alpha = 1.0$, $\beta = 0.6$, and $\gamma = 0.3$; grid-aware cropping alignment probability $p_{\text{align}} = 0.6$; and grid size $L = 4$ (i.e., 4×4 grids), and $r = 32$. These values were selected based on preliminary validation and are kept constant across datasets to ensure fairness and reproducibility.

Efficiency and scalability The proposed single-step enhancement introduces only a **16.9%** one-time overhead (3m17s enhancement vs. 16m05s training), amortized across epochs, preserving the efficiency profile of prior distillation methods. SSDIM has runtimes (∼9s, ∼15s, ∼40s,∼70s), for (100, 200, 500, 1k) images, resp on A6000 GPU.

3.3. Comparisions with State of the Art

Woof Experiments

We evaluate our proposed **Grid-Distil** framework on the ImageNet-Woof dataset under varying image-per-class (IPC) regimes, ranging from 10 (0.8%) to 100 (7.7%) samples per class, and across three architectures: ConvNet-6, ResNetAP-10, and ResNet-18. Table 1 compares our results against a comprehensive suite of recent dataset distillation and condensation methods, including K-Center, Herding, DiT, DM, IDC-1, GLaD, Minimax, D^4M, and VLCP.

Across all IPC settings and architectures, **Grid-Distil consistently achieves the highest accuracy**, surpassing prior state-of-the-art approaches by large margins. Under the extremely low data regime (IPC = 10), our method attains **49.7%** with ConvNet-6 and **65.5%** with ResNet-18, outperforming the next best VLCP baseline by over **15 percentage points**. As the data ratio increases, the gains persist: at IPC = 50, **Grid-Distil** reaches **84.3%** on ResNet-18, while VLCP remains at 58.9%, highlighting the robustness of our distilled representations.

ImageNette and ImageIDC: We further benchmark our **Grid-Distil** framework on the ImageNette and ImageIDC

Method	IPC=10	IPC=20	IPC=50
Random	54.2±1.6	63.5±0.5	76.1±1.1
DiT	59.1±0.7	64.8±1.2	73.3±0.9
DM	60.8±0.6	66.5±1.1	76.2±0.4
Minimax	57.7±1.2	64.7±0.8	73.9±0.3
D^4M	60.9±1.7	66.3±1.3	77.7±1.1
VLCP	64.8±3.6	71.4±0.5	81.2±0.8
Grid-Distil (Ours)	**83.3±0.6**	**87.3±0.5**	**92.7±0.7**

Table 2. Comparison of dataset distillation methods on **ImageNette** using ResNetAP-10 under various IPC settings. Best results are in bold; second-best method (VLCP) is underlined.

Method	IPC=10	IPC=20	IPC=50
Random	48.1±0.8	52.5±0.9	68.1±0.7
DiT	54.1±0.4	58.9±0.2	64.3±0.6
DM	52.8±0.5	58.5±0.4	69.1±0.8
Minimax	51.9±1.4	59.1±3.7	69.4±1.4
D^4M	50.3±1.0	55.8±0.2	69.1±2.4
VLCP	57.0±1.4	63.3±1.2	71.9±0.4
Grid-Distil (Ours)	**73.5±1.5**	**79.7±0.7**	**87.1±0.5**

Table 3. Comparison of dataset distillation methods on **ImageIDC** using ResNetAP-10 under various IPC settings. Best results are in bold; second-best method (VLCP) is underlined.

datasets to assess its generalization capability across diverse visual domains and semantic granularities. Both datasets represent compact yet challenging subsets of ImageNet, emphasizing robustness in low-shot settings. Tables 2 and 3 summarize results under multiple IPC regimes (10, 20, and 50) using a unified ResNetAP-10 evaluation protocol.

Across all IPC configurations, **Grid-Distil** achieves substantial performance gains over existing dataset distillation methods. On **ImageNette**, our approach reaches **83.3%**, **87.3%**, and **92.7%** accuracy for IPC 10, 20, and 50 respectively—surpassing the second-best VLCP by margins exceeding **11–20 percentage points**. Similarly, on the more complex **ImageIDC** dataset, **Grid-Distil** delivers consistent improvements, achieving **73.5%**, **79.7%**, and **87.1%**, again outperforming VLCP by a wide margin across all data scales.

Results on ImageNet-1K To evaluate the scalability of our **Grid-Distil** framework to large-scale and semantically diverse datasets, we conduct experiments on the challenging **ImageNet-1K** benchmark. We follow the **RDED protocol** and utilize the open-source implementation of **VLCP** for evaluation to ensure a fair and standardized comparison. All methods, including ours, are trained for the same number of epochs and under identical hyperparameter configurations as specified in the VLCP public codebase. The distilled sets are evaluated using a **ResNet-18** backbone trained on the synthetic data.

Table 4 summarizes our results at IPC = 10. Prior state-of-the-art methods such as Minimax and VLCP achieve accuracies of 44.3% and 46.7%, respectively. Our **Grid-Distil** framework attains a notable improvement, reaching **50.01%** accuracy when combined with our proposed *detail enhancement* strategy, surpassing VLCP by a clear margin while maintaining competitive variance. The bilinear variant of our method (without detail enhancement) already performs comparably to diffusion-based distillers, demonstrating the strength of our grid-structured submodular selection alone. Importantly, the consistent improvement under a rigorous, publicly reproducible setup underscores the robustness and generalizability of **Grid-Distil** when scaled to complex visual domains such as ImageNet-1K.

Method	Venue	Mean	Std
SRe^2L	NeurIPS'23	21.3	0.6
RDED	CVPR'24	42.0	0.1
DiT	CVPR'24	39.6	0.4
Minimax	CVPR'24	44.3	0.5
VLCP	ICCV'25	46.7	0.4
Ours (Bilinear)	–	35.40	0.25
Ours (Detail Enhancement)	–	**50.01**	0.29

Table 4. **Performance comparison on ImageNet-1K (IPC = 10).** Our diffusion-based detail enhancement significantly improves classification accuracy compared to bilinear preprocessing and recent distillation methods.

3.4. Ablative Study of Components

To better understand the effectiveness of our Grid Distillation pipeline, we conduct complementary ablation studies focusing on: (i) the role of **Diffusion-Based Detail Enhancement** in visual preprocessing, (ii) the influence of submodular optimization parameters (α, β, γ) governing coverage, diversity, and spectral consistency in our grid selection formulation and iii) incorporation of semantic information in the prompts.

Effect of Detail Enhancement. Table 5 presents a comparative evaluation between **Bilinear Upsampling** and our proposed **Diffusion-Based Detail Enhancement** across ImageIDC, ImageNette, and ImageWoof datasets at varying IPC levels. All experiments utilize the ResNetAP-10 backbone under identical training conditions.

We observe that our enhancement mechanism consistently improves recognition accuracy, particularly for datasets exhibiting high intra-class variability such as ImageNette, ImageWoof and ImageNet-1K. The improvement stems from the diffusion prior's ability to restore fine spatial frequencies lost during naive resampling, thereby generating more faithful edge and texture representations.

Dataset	Bilinear Upsampling			Detail Enhancement (Ours)		
	IPC 10	IPC 20	IPC 50	IPC 10	IPC 20	IPC 50
ImageIDC	74.7±0.2	79.8±1.3	87.7±0.5	73.5±1.5	79.7±0.7	87.1±0.5
ImageNette	81.5±0.9	85.7±1.0	91.5±0.5	**83.3**±0.6	**87.3**±0.5	**92.7**±0.7
ImageWoof	62.5±0.7	73.1±0.3	82.2±0.7	**63.9**±0.6	**73.7**±0.7	**82.4**±0.2

Table 5. **Ablation study on detail enhancement preprocessing.** Comparison between **Bilinear Upsampling** and our **Diffusion-Based Detail Enhancement** across three datasets using ResNetAP-10 at varying IPC values. Our preprocessing consistently preserves fine-grained visual detail, leading to improved classification performance on ImageNette and ImageWoof.

While ImageIDC shows marginal differences—likely due to its low-frequency visual composition—our approach offers clear gains (at IPC=10 on ImageNet) when fine-grained local details are critical to class discrimination.

Influence of Submodular Parameters (α, β, γ)**.** We further analyze the sensitivity of our Spectral-Submodular Grid objective to the relative weighting of its components. Specifically, α, β, and γ respectively control *coverage*, *diversity*, and *spectral consistency* among selected patches. The experiments are performed on real (non-distilled) ImageWoof grids at IPC=20, using a ResNetAP-10 backbone to isolate the contribution of submodular optimization.

Table 6. **Ablation of submodular ooptimization parameters.** Results on real ImageWoof grids (IPC=20) using ResNetAP-10. The study isolates the role of coverage (α), diversity (β), and spectral information (γ).

ID	α	β	γ	Top1 Acc	Description
-	0.0	0.0	0.0	74.6	Random
1	0.0	0.0	1.0	75.8	Only γ active
2	0.0	1.0	0.0	74.4	Only β active
3	0.3	0.6	1.0	76.0	γ-dominant
4	0.3	1.0	0.6	74.6	β-dominant
5	0.6	0.3	1.0	73.0	γ-dominant
6	0.6	0.6	0.6	75.6	Balanced regime
7	0.6	1.0	0.3	77.8	β-dominant
8	1.0	0.0	0.0	78.2	Only α active
9	1.0	0.3	0.6	78.0	α-dominant
10	1.0	0.6	0.3	**78.6**	α-dominant

Prompt Ablation on UCF-101 Mid-Frames To further investigate the role of semantic conditioning in our framework, we conduct an ablation on **UCF-101 Mid-Frames** by incorporating *action labels* into the grid enhancement prompts. Following the few-shot evaluation protocol of **LDC** [17], we employ a ResNet-50 backbone and assess performance across 1-, 2-, 4-, and 8-shot settings using the official implementation. As shown in the results, introducing action labels (**Grid-LDC-Labels**) consistently improves accuracy over the base **Grid-LDC** variant across all shot regimes. This gain highlights that semantically guided enhancement provides a richer inductive bias for action understanding, improving representation alignment between distilled images and downstream recognition tasks.

Method	1-shot	2-shot	4-shot	8-shot
CLIP (ZS) [28]	61.78	–	–	–
CoOp [52]	62.67	63.60	69.40	72.50
Proto-CLIP [26]	63.15	67.46	69.50	71.08
Proto-CLIP-F [26]	63.07	67.49	69.55	74.81
CLIP-Adapter [7]	62.77	64.17	69.10	72.90
SuS-X [36]	62.78	66.45	66.87	68.88
APE	63.26	65.72	69.92	71.74
Tip-Adapter [46]	62.73	64.76	66.19	68.46
Tip-Adapter-F [46]	64.95	66.38	70.98	74.65
LDC [17]	66.83	69.81	74.12	77.03
Ours (Grid-LDC)	71.66	75.02	75.87	78.01
Ours (Grid-LDC-Labels)	**73.88**	**75.65**	**77.37**	**79.03**

Table 7. **Few-shot classification accuracy (%) on UCF-101 Mid-Frames.** Our method (Grid-LDC) consistently outperforms prior CLIP adaptation techniques across all few-shot settings, demonstrating the benefit of grid-level distillation for motion-centric visual reasoning.

4. Conclusion

We introduced **Grid Distillation**, a generative dataset distillation framework that compresses large datasets into compact, informative grids. By formulating the selection as a **spectral submodular optimization** problem, our approach injects *world knowledge* from CLIP to maximize coverage, diversity, and spectral fidelity. A diffusion-based **detail enhancement** step restores natural image statistics, while a grid-aware cropping strategy improves discriminability without altering standard inference pipelines. Across ImageWoof, ImageNette, ImageIDC, and ImageNet-1K, our method consistently outperforms prior distillation approaches, establishing a new direction for knowledge-guided dataset compression.

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